

Preliminaries

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Objective

Use logic, sets, quantifiers, and proof methods to follow and construct mathematical arguments.

- ◆ Identify a statement or a predicate.
- ◆ Translate among logical notation, set notation, quantified language, and ordinary language.
- ◆ Negate and evaluate compound or quantified statements.
- ◆ Know direct proof, contraposition, induction, or counterexample.

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Section 1

Propositional logic

Statements

Definition 1. A **statement** is a declarative sentence that, once the meanings of its terms and the context are fixed, is either true or false. (Its truth value need not be known.)

EXAMPLES

- ✓ $2 + 3 = 5$.
- ✓ $2 + 3 = 6$. (False, but still a statement.)
- ✓ Every even integer greater than 2 is the sum of two primes. (A statement; whether it is true or false is not currently known.)
- ✗ " $x + 2 > 5$ " is not a statement as written, because its truth value depends on the unspecified value of x .

Is it a statement?

Assume all named objects and contextual references are fixed.

- a** Seven is a prime number.
- b** Every real number is positive.
- c** If 12 is even, then 12^2 is even.
- d** This rule is best.
- e** There exists a real number whose square is 2.
- f** Today is warmer than yesterday.
- g** Every person in the room has a ticket.
- h** $x + 2 > 5$.

Statements and expressions with a variable

Statements. Items (a)–(g) are statements once the named objects, comparison criterion, place, date, and group of people have been fixed. Each then has a truth value, even when that truth value is unknown.

Item (b) is false and remains a statement because it is truth-valued.

An expression with a variable. Item (h),

$$x + 2 > 5,$$

has no truth value until x is pinned down. It becomes a statement once a value is substituted for x , or once x is quantified over a specified domain. Section 3 names these objects and treats them systematically.

Video: [Mathematical Visual Proofs, Predicates and quantifiers](#).

Compound statements

Definition 2. A **compound statement** is a statement built from one or more statements using logical operators (connectives).

If P and Q are statements, then each of

$$\neg P, \quad P \wedge Q, \quad P \vee Q, \quad P \Rightarrow Q, \quad P \Leftrightarrow Q$$

is a compound statement. The next slides give the truth conditions for these operators.

Negation

If a statement P is true, its negation $\neg P$ is false, and vice versa. Negation simply reverses the truth value.

P	$\neg P$
T	F
F	T

Table 1 · Negation

Conjunction & disjunction

Conjunction $P \wedge Q$: “and”. True only when both are true.

Disjunction $P \vee Q$: “or”. True when at least one is true; false only when both fail.

In mathematics, $\vee$ is **inclusive**: it is also true when both statements are true.

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Table 2 · Conjunction & disjunction

Exclusive or and symmetric difference

Two versions of “exactly one”

For statements P and Q , exclusive or is

$$(P \vee Q) \wedge \neg(P \wedge Q).$$

It is true when exactly one statement is true.

Further reading: [Osborne, *Mathematical Methods for Economic Theory*](#), logic and sets.

For sets A and B , symmetric difference is

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

It contains the elements in exactly one set.

Videos: [Neso Academy, *Exclusive OR*](#); [MathsSmart, *Symmetric difference*](#).

Implication

$P \Rightarrow Q$: “if P , then Q ”.

P is the **antecedent** (often the hypothesis), and Q is the **consequent** (often the conclusion).

Take $P = “12 > 10”$, $Q = “12 > 8”$.

P true, Q true $\Rightarrow$ true.

P true, Q false $\Rightarrow$ false.

P false $\Rightarrow$ true regardless of Q .

Equivalent form: $P \Rightarrow Q \Leftrightarrow \neg P \vee Q$.

Vacuous truth. If P is false, then $P \Rightarrow Q$ is true regardless of Q .

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Why the false row gives a counterexample

From $P \Rightarrow Q$ alone, **neither P nor Q follows**. If P is also given, then Q follows.

An example

EXAMPLE

P = "Trump was born in outer space" (false),
 Q = "I was born in China".

Since P is false, $\neg P$ is true. Hence

$$P \Rightarrow Q \text{ is true.}$$

We do not need to know whether Q is true.

WHAT THE EXAMPLE SHOWS

1 • False antecedent: vacuous truth.

A false P alone makes $P \Rightarrow Q$ true.

We do not need to know whether Q is true.

2 • No causation required.

$P \Rightarrow Q$ is fixed by the truth values of P and Q alone, so it asserts no causal link between them.

Equivalence

$P \Leftrightarrow Q$: **“if and only if”**. It is true when P and Q share a truth value and false when they differ.

It combines two implications:

$$(P \Rightarrow Q) \wedge (Q \Rightarrow P).$$

Equivalently, it is true exactly when both statements are true or both are false:

$$(P \wedge Q) \vee (\neg P \wedge \neg Q).$$

This is the negation of exclusive or.

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautology, contradiction, contingency

TAUTOLOGY

True under every assignment of truth values.

$$P \vee \neg P$$

CONTRADICTION

False under every assignment of truth values.

$$P \wedge \neg P$$

CONTINGENT

True under some assignments and false under others.

$$P \wedge Q$$

Two formulas are **logically equivalent** if they have the same truth value under every assignment. Equivalently, joining them with $\Leftrightarrow$ produces a tautology. For implication,

$$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$$

is a tautology. Hence $P \Rightarrow Q$ and $\neg P \vee Q$ are logically equivalent.

Video: [TrevTutor, Tautologies, contradictions, and contingencies.](#)

Negating compound statements

De Morgan's laws describe how negation passes through “and” and “or”.

NEGATING A CONJUNCTION

$$\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q).$$

“Not both” means that at least one fails.

NEGATING A DISJUNCTION

$$\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q).$$

“Neither” means that both fail.

Converse & contrapositive

Definition 3. Given a statement $P \Rightarrow Q$:

Converse

$$Q \Rightarrow P$$

Even when $P \Rightarrow Q$ is true, $Q \Rightarrow P$ can be false.

Example: "If n is divisible by 4, then n is even" is true; its converse fails at $n = 2$.

Contrapositive

$$\neg Q \Rightarrow \neg P$$

In every row of the truth table, $\neg Q \Rightarrow \neg P$ has the same truth value as $P \Rightarrow Q$.

Use this equivalence in a proof by contraposition

Sufficient & necessary conditions

P sufficient for Q $P \Rightarrow Q.$

P necessary for Q $Q \Rightarrow P.$

P necessary and sufficient $P \Leftrightarrow Q.$

Video: Dr. Trefor Bazett, *Necessary and sufficient conditions*.

Operators in practice

1 · $P = \text{"12 IS EVEN"} , Q = \text{"12}^2 \text{ IS EVEN"}$

Write the negation of P .

Write $P \wedge Q$ and $P \vee Q$.

Write $P \Rightarrow Q$; explain what it means.

Write $P \Leftrightarrow Q$; when is it true?

2 · EVALUATE WITH A TRUTH TABLE

$$P \wedge (P \vee Q)$$

$$\neg(P \Rightarrow Q)$$

$$(P \Rightarrow Q) \Rightarrow (\neg Q \Rightarrow \neg P)$$

Operators with two true statements

Here P is "12 is even" and Q is " 12^2 is even". Thus P is true and Q is true.

NEGATION, CONJUNCTION, DISJUNCTION

$\neg P$: "12 is not even" (false),

$P \wedge Q$: both statements hold (true),

$P \vee Q$: at least one holds (true).

IMPLICATION AND EQUIVALENCE

$P \Rightarrow Q$: If 12 is even, then 12^2 is even (true),

$P \Leftrightarrow Q$: P and Q share a truth value (true).

The equivalence is also true when both component statements are false.

Three truth-table calculations

P	Q	$P \wedge (P \vee Q)$	$\neg(P \Rightarrow Q)$	$(P \Rightarrow Q) \Rightarrow (\neg Q \Rightarrow \neg P)$
T	T	T	F	T
T	F	T	T	T
F	T	F	F	T
F	F	F	F	T

$P \wedge (P \vee Q) \Leftrightarrow P$. The formula takes the value of P .

$\neg(P \Rightarrow Q) \Leftrightarrow P \wedge \neg Q$. The last formula is a tautology because its consequent is the contrapositive of its antecedent.

Threshold conditions

Let $m \geq 0$, and define Q by the condition $m \geq 8$.

Let P be " $m \geq 10$ " and let Q be " $m \geq 8$." Is P sufficient for Q ? Is P necessary for Q ?

Express each answer as an **implication** and justify any negative answer with a **counterexample**.

Threshold conditions

Let P mean $m \geq 10$ and Q mean $m \geq 8$.

SUFFICIENCY

If $m \geq 10$, then $m \geq 8$. Hence

$$P \Rightarrow Q,$$

so P is sufficient for Q .

NECESSITY

At $m = 9$, Q is true and P is false. Thus $Q \Rightarrow P$ fails, so P is not necessary for Q .

Section 2

Set theory

Sets, membership, and inclusion

Definition 4. A **set** S is a collection of objects, called its **elements**.

BASIC NOTATION

Membership $x \in S$ x is an element of S .

Non-membership $x \notin S$ x is not an element of S .

Subset $T \subseteq S$ Every element of T is also in S .

Proper subset $T \subsetneq S$ $T \subseteq S$ and $T \neq S$.

Superset $S \supseteq T$ Equivalently, $T \subseteq S$.

ELEMENTWISE TESTS

Inclusion is an implication applied to every element:

$$A \subseteq B \Leftrightarrow (x \in A \Rightarrow x \in B) \text{ for every } x.$$

Equality means membership agrees in both directions:

$$A = B \Leftrightarrow (x \in A \Leftrightarrow x \in B) \text{ for every } x.$$

Equivalently,

$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A.$$

Membership and inclusion are different: $x \in S$ compares an object with a set, whereas $A \subseteq S$ compares two sets.

The next section writes “for every x ” as $\forall x$.

Video: Dr. Trefor Bazett, *Intro to sets: examples, notation and properties*.

Empty set and power set

Empty set $\emptyset$ The unique set with no elements.

Power set $\mathcal{P}(S)$ The set of all subsets of S ; in particular, $\emptyset, S \in \mathcal{P}(S)$.

For example, if $S = \{a, b\}$, then

$$\mathcal{P}(S) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

For a finite set, $|S|$ denotes its number of elements. Hence, if $|S| = n$,

$$|\mathcal{P}(S)| = 2^n.$$

The symbols $\subset$ **and** $\supset$ may include equality or denote proper inclusion, depending on the author.

Standard number sets

$\mathbb{N}$ $\{1, 2, 3, \dots\}$.

$\mathbb{Z}$ Integers: $\dots, -2, -1, 0, 1, 2, \dots$

$\mathbb{Q}$ Rational numbers: fractions of integers with nonzero denominator.

$\mathbb{R}$ Real numbers.

$\mathbb{R}_+$ In these slides, $[0, \infty)$; $\mathbb{R}_{++} = (0, \infty)$.

In these slides, $\mathbb{N} = \{1, 2, 3, \dots\}$; use $\mathbb{Z}_{\geq 0}$ for the nonnegative integers. Some authors include 0 in $\mathbb{N}$.

Intervals in $\mathbb{R}$

For real numbers $a < b$, endpoint brackets show whether the endpoints are included:

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} : a < x < b\},$$

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}, \quad (a, b] = \{x \in \mathbb{R} : a < x \leq b\}.$$

Unbounded intervals are written, for example,

$$(-\infty, b), \quad (-\infty, b], \quad (a, \infty), \quad [a, \infty).$$

$\pm\infty$ are not real endpoints, so an infinite end always uses a parenthesis. In particular, $\mathbb{R}_+ = [0, \infty)$ and $\mathbb{R}_{++} = (0, \infty)$ in these slides.

Video: [Socratica, How to use interval notation](#).

Set-builder notation

Given a set X and a condition $P(x)$ that each element either satisfies or fails, the elements satisfying it can be written in set-builder notation:

$$S = \{x \in X : P(x)\}.$$

Thus a condition on the elements of X determines a subset of X : S is the set of points at which $P(x)$ is true. Section 3 studies such conditions as **predicates**.

For example,

$$[0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}.$$

Within set-builder notation, “:” or “|” means “such that”.

Sets and n -tuples

A **set** fixes *which* elements are present, but **not their order or multiplicity**:

$$\{a, b\} = \{b, a\}, \quad \{a, a, b\} = \{a, b\}.$$

An **n -tuple** is an ordered list of n entries,

$$(x_1, \dots, x_n).$$

Its i th entry x_i is the i th **component** (or coordinate). **Order matters and repetitions are allowed**:

$$(a, b) \neq (b, a) \text{ if } a \neq b, \quad (a, a, b) \text{ is a valid 3-tuple.}$$

Two n -tuples are equal exactly when all corresponding components are equal:

$$(x_1, \dots, x_n) = (y_1, \dots, y_n) \Leftrightarrow x_i = y_i \text{ for every } i = 1, \dots, n.$$

Definition notation

A definition introduces a new object or fixes notation. Three forms are in common use:

$x := y$ x is defined as y .

$x \stackrel{\text{def}}{=} y$ The word “definition” is written above the equals sign.

$x \triangleq y$ A third symbol for the same definition relation.

Some authors also use $x \equiv y$ for definitions, but $\equiv$ may also mean identity, logical equivalence, or congruence.

Cartesian products

Definition 5. For sets A and B , their **Cartesian product** and its membership test are

$$A \times B := \{(a, b) : a \in A \text{ and } b \in B\},$$

$$(a, b) \in A \times B \Leftrightarrow a \in A \text{ and } b \in B.$$

Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$. Then

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}.$$

Hence $|A \times B| = |A| |B| = 2 \cdot 3 = 6$. Moreover, $(a, 1) \in A \times B$ but $(a, 1) \notin B \times A$, so in general $A \times B \neq B \times A$.

Video: [TrevTutor, Cartesian products and ordered pairs.](#)

Finite products and $\mathbb{R}^n$

For sets $A_1, \dots, A_n$,

$$A_1 \times \cdots \times A_n := \{(x_1, \dots, x_n) : x_i \in A_i \text{ for every } i\},$$

$$(x_1, \dots, x_n) \in A_1 \times \cdots \times A_n \Leftrightarrow x_i \in A_i \text{ for every } i.$$

If every factor is the same set A , write

$$A^n := \underbrace{A \times \cdots \times A}_{n \text{ times}}.$$

If the factors are finite, then

$$|A_1 \times \cdots \times A_n| = \prod_{i=1}^n |A_i|.$$

Two product spaces used throughout the course are

$$\mathbb{R}^n = \{(x_1, \dots, x_n) : x_i \in \mathbb{R} \text{ for every } i\},$$

$$\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0 \text{ for every } i\}.$$

Consumption and price vectors

Bundles, prices, and budgets

A consumption bundle and a price vector are

$$x = (x_1, \dots, x_n) \in \mathbb{R}_+^n,$$

$$p = (p_1, \dots, p_n) \in \mathbb{R}_{++}^n.$$

Coordinate i refers to good i . Thus, for two goods, $(3, 1)$ and $(1, 3)$ generally describe different bundles.

For income $m \geq 0$, expenditure on x and the corresponding budget set are

$$p \cdot x = \sum_{i=1}^n p_i x_i,$$

$$B(p, m) = \{x \in \mathbb{R}_+^n : p \cdot x \leq m\}.$$

Set operations

Each operation forms a new set. Complements are taken relative to a fixed universe U .

Complement

$$A^c = U \setminus A$$

Elements in U but not in A .

Union

$$A \cup B$$

Elements in A or B , including those in both.

Intersection

$$A \cap B$$

Elements in both A and B .

Difference

$$A \setminus B$$

Elements in A but not in B .

Video: [TrevTutor, Set Operations](#).

Set operations

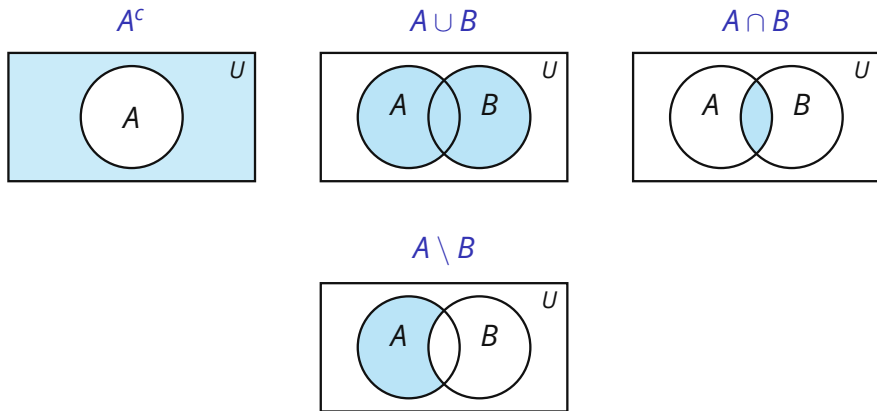


Figure 1 · Each shaded region is the set named above it.

Properties

All complements are taken relative to the same fixed universe U .

Distributive

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Absorption

$$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$$

De Morgan's laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Complement

$$A \cup A^c = U, \quad A \cap A^c = \emptyset$$

Under elementwise membership, $\cup$ acts like $\vee$, $\cap$ like $\wedge$, and complement like $\neg$. Hence the set-theoretic De Morgan laws mirror the logical ones.

Sets versus tuples

Decide whether each statement is true or false.

a $\{1, 2\} = \{2, 1\}$

b $\{1, 1, 2\} = \{1, 2\}$

c $(1, 2) = (2, 1)$

d $(1, 1, 2) \in \{1, 2\}^3$

e $\{1, 2\} \in \mathbb{R}^2$

f $(1, 2) \in \mathbb{R}^2$

Sets and ordered tuples

- a* True. A set ignores order.
- b* True. Repetition does not create a new set element.
- c* False. Tuple coordinates are ordered.

- d* True. Every coordinate lies in $\{1, 2\}$.
- e* False. Elements of $\mathbb{R}^2$ are ordered pairs.
- f* True. $(1, 2)$ is an ordered pair of reals.

Set identities

Let A, B, C be subsets of a universal set X . Show that:

a $A \setminus B = B \setminus A \Leftrightarrow A = B$

b $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

c $(B \cup C)^c = B^c \cap C^c$ **(De Morgan)**

Video: [Wrath of Math, Proof of De Morgan's laws for set complement.](#)

Equality of differences

Suppose $A \setminus B = B \setminus A$. If $x \in A \setminus B$, the equality gives $x \in B \setminus A$, which would require both $x \in A$ and $x \notin A$. Hence $A \setminus B = \emptyset$, so $A \subseteq B$. By the same argument, $B \setminus A = \emptyset$, so $B \subseteq A$. Therefore $A = B$.

Conversely, if $A = B$, then

$$A \setminus B = A \setminus A = \emptyset = B \setminus B = B \setminus A.$$

Thus

$$A \setminus B = B \setminus A \iff A = B.$$

Membership proofs for two identities

For any $x \in X$,

$$\begin{aligned}x \in A \cap (B \cup C) &\Leftrightarrow x \in A \wedge (x \in B \vee x \in C) \\&\Leftrightarrow (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\&\Leftrightarrow x \in (A \cap B) \cup (A \cap C).\end{aligned}$$

Likewise,

$$\begin{aligned}x \in (B \cup C)^c &\Leftrightarrow x \notin B \cup C \\&\Leftrightarrow x \notin B \wedge x \notin C \\&\Leftrightarrow x \in B^c \cap C^c.\end{aligned}$$

Since these equivalences hold for every x , the corresponding sets are equal.

Section 3

Quantifiers

Predicates and free variables

A **predicate** such as $P(x)$ has one or more free variables and **has** a truth value once all its free variables have been quantified or assigned values. A **statement** has no free variables and has a truth value. Formally, a formula with a free variable is **open**; one with none is **closed**, or a **statement**.

A FREE VARIABLE

For $x \in \mathbb{R}$, consider

$$P(x) : x + 2 > 5.$$

As written, x is free, so $P(x)$ is a predicate in x .

Substitute $P(4)$ is true; $P(0)$ is false.

Quantify $\forall x \in \mathbb{R}$, $P(x)$ and $\exists x \in \mathbb{R}$ s.t. $P(x)$ are statements.

Video: [TrevTutor, Scope, free and bound variables in predicate logic.](#)

What binds x ?

$$\forall x \in \mathbb{R}, x^2 \geq 0$$

statement: x is bound,

$$x \in \mathbb{R} \Rightarrow x^2 \geq 0$$

predicate in x : x is free,

$$\forall x \in \mathbb{R}, x < y$$

predicate in y : y is free,

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ s.t. } x < y$$

statement: both bound.

Writing $x \in \mathbb{R}$ states a condition; only $\forall x$ or $\exists x$ binds x . Thus $x^2 \geq 0$ is open when x is free.

Quantifiers over a set

Once a domain X and a property $P(x)$ are fixed, quantifiers specify how the property applies to elements of X .

$\exists$ **There exists.** $\exists x \in X$ s.t. $P(x)$ means that at least one $x \in X$ satisfies P .

$\exists!$ **There exists a unique.** $\exists! x \in X$ s.t. $P(x)$ means that exactly one $x \in X$ satisfies P .

$\forall$ **For all.** $\forall x \in X$, $P(x)$ means that every $x \in X$ satisfies P .

Video: Kimberly Brehm, *Quantifiers*.

Predicates, sets, and quantifiers

Fix a domain X and a predicate $P(x)$. Its **truth set** is

$$S_P := \{x \in X : P(x)\}.$$

QUANTIFIERS DESCRIBE THE TRUTH SET

$$\forall x \in X, P(x) \Leftrightarrow S_P = X,$$

$$\exists x \in X \text{ s.t. } P(x) \Leftrightarrow S_P \neq \emptyset,$$

$$\exists! x \in X \text{ s.t. } P(x) \Leftrightarrow S_P \text{ has exactly one element.}$$

LOGICAL RELATIONS BECOME SET RELATIONS

For another predicate $Q(x)$,

$$S_{\neg P} = X \setminus S_P,$$

$$S_{P \wedge Q} = S_P \cap S_Q,$$

$$S_{P \vee Q} = S_P \cup S_Q,$$

$$\forall x \in X, P(x) \Rightarrow Q(x) \Leftrightarrow S_P \subseteq S_Q,$$

$$\forall x \in X, P(x) \Leftrightarrow Q(x) \Leftrightarrow S_P = S_Q.$$

Existence, counterexamples, and the domain

The quantifiers also tell us what kind of evidence establishes or refutes a claim.

UNIVERSAL CLAIM

$$\forall x \in X, P(x)$$

To **disprove** it, one **counterexample** $x^* \in X$ with $\neg P(x^*)$ is enough. For example,

$$\forall x \in \mathbb{R}, x^2 \geq x$$

is false because $x^* = 1/2$ gives $1/4 < 1/2$.

EXISTENTIAL CLAIM

$$\exists x \in X \text{ s.t. } P(x)$$

To **prove** it, one specific element $x^* \in X$ with $P(x^*)$ is enough. For example, over $X = \mathbb{R}$, choosing $x^* = 1/2$ establishes

$$\exists x \in X \text{ s.t. } 2x = 1.$$

The domain is part of the statement. The claim $\exists x \in X \text{ s.t. } 2x = 1$ is true when $X = \mathbb{R}$ and false when $X = \mathbb{Z}$.

Order matters

Let S be a set of people, C a set of courses, and $T(s, c)$ mean that person s takes course c .

$$\forall s \in S, \exists c \in C \text{ s.t. } T(s, c)$$

Every person takes at least one course.

$$\exists c \in C \text{ s.t. } \forall s \in S, T(s, c)$$

One course is taken by every person.

Swapping $\forall$ and $\exists$ changes the statement.

Video: [Kimberly Brehm, Nested quantifiers and negations.](#)

Negating quantified statements

Negating a quantified statement switches the quantifier and negates the property:

$$\neg(\forall x \in X, P(x)) \Leftrightarrow \exists x \in X \text{ s.t. } \neg P(x),$$

$$\neg(\exists x \in X \text{ s.t. } P(x)) \Leftrightarrow \forall x \in X, \neg P(x).$$

"Not everyone satisfies P " means "someone does not satisfy P ";
"nobody satisfies P " means "everyone fails to satisfy P ."

Video: [Dr. Trefor Bazett, *Negating universal and existential quantifiers.*](#)

Say it in words

Translate each quantified statement into ordinary language.

$$\forall x \in \mathbb{R}, x^2 \geq 0.$$

$$\exists y \in \mathbb{Z} \text{ s.t. } y^2 = 16.$$

$$\exists! z \in \mathbb{N} \text{ s.t. } z + 1 = 2.$$

Each statement names a domain, then a quantifier, then a property.

Three quantified statements

 $\forall$

Every real number has a nonnegative square.

 $\exists$

Some integer has square 16;
for example, $4^2 = 16$.

 $\exists!$

There is one natural number z satisfying $z + 1 = 2$, namely $z = 1$.

Section 4

Proof techniques

Logical form tells us what a proof must do

The form of the claim determines what counts as a proof or a refutation.

CLAIM	TO PROVE	TO DISPROVE
$P \Rightarrow Q$	Assume P and derive Q , or prove the contrapositive.	Give a case with P true and Q false.
$\forall x \in X, P(x)$	Fix an arbitrary $x \in X$ and prove $P(x)$.	Give one counterexample $x^* \in X$ with $\neg P(x^*)$.
$\exists x \in X$ s.t. $P(x)$	Give a specific $x^* \in X$ satisfying P .	Prove $\forall x \in X, \neg P(x)$.
$P \Leftrightarrow Q$	Prove both $P \Rightarrow Q$ and $Q \Rightarrow P$.	Refute at least one of the two implications.
$\exists! x \in X$ s.t. $P(x)$	Prove existence and then uniqueness.	Show no element satisfies P , or exhibit two distinct elements satisfying P .

Definitions used in examples

Several proof examples use the algebraic definitions of even and odd integers.

Even. An integer n is even if $n = 2k$ for some integer k .

Odd. An integer n is odd if $n = 2k + 1$ for some integer k .

1 · Direct proof

A direct proof starts from the assumptions and reaches the conclusion through valid steps.

The sum of two even integers is even.

Let $a = 2m$ and $b = 2n$ for some integers m, n . Then

$$a + b = 2m + 2n = 2(m + n).$$

Since $m + n$ is an integer, $a + b$ is even. ■

Common end-of-proof marks include $\square$, ■, Q.E.D., QED, #, and //.

2 · Contraposition

Recall: $P \Rightarrow Q \Leftrightarrow \neg Q \Rightarrow \neg P$. It is sometimes easier to assume $\neg Q$ and derive $\neg P$.

Claim. If n^2 is even, then n is even.

Contrapositive: if n is odd, then n^2 is odd.

If n is odd, then $n = 2k + 1$ for some integer k . Therefore

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. Hence, by contraposition, if n^2 is even, then n is even. ■

3 · Contradiction

To prove P by contradiction, assume $\neg P$ and derive a contradiction. Since a contradiction has the form $Q \wedge \neg Q$, the logical pattern is

$$(\neg P \Rightarrow (Q \wedge \neg Q)) \Rightarrow P.$$

This method is often useful for impossibility, nonexistence, irrationality, and uniqueness arguments.

There is no largest real number.

Suppose, for contradiction, that M is the largest real number. Then $M + 1$ is also a real number, and $M + 1 > M$. This contradicts the assumption that M is largest. Therefore no largest real number exists. ■

4 · Existence & uniqueness

To prove $\exists x \in X$ s.t. $P(x)$, it is often enough to give one concrete $x^* \in X$ with $P(x^*)$ true.

To prove $\exists! x \in X$ s.t. $P(x)$, prove two claims:

- 1 Existence: at least one $x \in X$ satisfies $P(x)$.
- 2 Uniqueness: if $x, y \in X$ both satisfy P , then $x = y$.

5 · Counterexample

To disprove a universal claim, it is enough to find one case in which it fails.

Recall the full truth table

Claim. If $x^2 > 4$, then $x > 2$.

This claim is false. Take $x = -3$. Then $x^2 = 9 > 4$, but $x > 2$ is false. A single example with true premise and false conclusion disproves the claim. ■

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The single false row Only a true premise and a false conclusion make $P \Rightarrow Q$ false:

$$\neg(P \Rightarrow Q) \Leftrightarrow P \wedge \neg Q.$$

This is why one such case disproves a universal implication.

6 · Induction

INTUITION

A **chain reaction**: establish a starting case, then show that each case forces the next. It proves a universal claim over an integer tail, such as

$$\forall n \in \mathbb{Z}, \quad n \geq n_0 \Rightarrow P(n).$$

PRINCIPLE

Let n_0 be an integer. To prove $P(n)$ for every integer $n \geq n_0$, establish:

Base case Prove $P(n_0)$.

Inductive step Fix an arbitrary integer $k \geq n_0$.
Assume $P(k)$ and prove $P(k+1)$.

Then $P(n)$ holds for every integer $n \geq n_0$.

Video: [Learn Math Tutorials, Proof by mathematical induction.](#)

Induction at work

Prove, for every integer $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Base $n = 1$: $1 = 1(1+1)/2$.

Step: assume $1 + 2 + \cdots + k = k(k+1)/2$. Then

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Hence the formula holds for every $n \geq 1$. ■

Gibbard-Satterthwaite: the base case

Base case in the alternatives dimension

Gibbard-Satterthwaite theorem. Let $m = |A|$. In the classical unrestricted-preference setting, if $m \geq 3$, every onto strategy-proof social choice function is dictatorial.

THE BASE CASE NEED NOT BE 1

The theorem itself is stated for $m \geq 3$. Suppressing the other parameter, write $Q(m)$ for the claim in the alternatives dimension.

$$Q(3), \quad Q(m) \Rightarrow Q(m+1) \quad (m \geq 3).$$

No proof of $Q(1)$ or $Q(2)$ is needed.

Gibbard–Satterthwaite: two induction dimensions

Induction across agents and alternatives

Write $P(n, m)$ for the claim with n agents and m alternatives. The Schmeidler–Sonnenschein route starts from the two-agent, three-alternative case and then extends in both integer parameters.

STARTING CASE AND INDUCTION DIRECTIONS

Starting case $P(2, 3)$

$n : 2 \rightarrow 3 \rightarrow 4 \rightarrow \dots$

$m : 3 \rightarrow 4 \rightarrow 5 \rightarrow \dots$

Induction can be nested across integer dimensions.

Source: Barberà, *Strategyproof Social Choice*, §3.3.2; Schmeidler–Sonnenschein (1978).

Prove or give a counterexample

1. If $x > 3$ and $y > 2$, then $x + y > 5$.
2. If $n \in \mathbb{Z}$ and n^2 is odd, then n is odd.
3. For every $N \in \mathbb{N} \cup \{0\}$ and $r \in \mathbb{R} \setminus \{1\}$,
4. For every integer $n \geq 3$, $2^n > 2n + 1$.
5. If $x < y$, then $x^2 < y^2$.
6. For real x, y , $x^2 = y^2 \Leftrightarrow x = y$.

$$\sum_{i=0}^N r^i = \frac{r^{N+1} - 1}{r - 1}.$$

Statements 1–2

1 • DIRECT ARGUMENT

From $x > 3$ and $y > 2$, addition preserves strict inequalities:

$$x + y > 3 + 2 = 5.$$

Hence the statement is true.

2 • CONTRAPOSITIVE

Prove the contrapositive. If n is even, $n = 2k$ for some $k \in \mathbb{Z}$. Then

$$n^2 = 4k^2 = 2(2k^2),$$

which is even. Therefore n^2 odd implies n odd.

Statements 3-4

3 · GEOMETRIC SUM

For $N = 0$,

$$1 = \frac{r-1}{r-1}.$$

Assume $\sum_{i=0}^k r^i = (r^{k+1} - 1)/(r - 1)$. Then

$$\begin{aligned} \sum_{i=0}^{k+1} r^i &= \frac{r^{k+1} - 1}{r - 1} + r^{k+1} \\ &= \frac{r^{k+2} - 1}{r - 1}. \end{aligned}$$

Hence the identity holds by induction.

4 · EXPONENTIAL BOUND

At $n = 3$, $2^3 = 8 > 7$. Assume $2^k > 2k + 1$, with $k \geq 3$. Then

$$2^{k+1} > 4k + 2 > 2k + 3 = 2(k + 1) + 1,$$

because $2k - 1 > 0$. Thus the claim follows by induction.

Statements 5–6

Both statements are false.

For Statement 5, take

$$x = -2, \quad y = -1.$$

Then $x < y$, but $x^2 = 4 > 1 = y^2$.

For Statement 6, take $x = 1$ and $y = -1$. Then $x^2 = y^2 = 1$, but $x \neq y$. Factoring gives the correct equivalence:

$$\begin{aligned} x^2 = y^2 &\Leftrightarrow x^2 - y^2 = 0 \\ &\Leftrightarrow (x - y)(x + y) = 0 \\ &\Leftrightarrow x = y \text{ or } x = -y. \end{aligned}$$

Video: [Dr Trefor Bazett, *Disproving implications with counterexamples.*](#)

How results are named

Result labels as dependency roles

Theorem Often reserved for one of a paper's broad main results.

Proposition A theorem-like result; often used for a more local or narrower result, though the distinction from theorem is conventional.

Lemma A result used mainly to establish another result.

Corollary A direct consequence or special case of an earlier result.

Theorem and proposition have no fixed logical ranking; usage depends on the paper's organization and the author's conventions.

Video: Quoc Dat Phung, *What is a theorem, corollary, conjecture, lemma, axiom.*

How theory papers use result labels

Result labels in published papers

THREE EXAMPLES FROM LEADING ECONOMICS JOURNALS

AER, Kamenica and Gentzkow (2011). *Bayesian Persuasion* calls its basic characterization Proposition 1 and an immediate implication Corollary 1.

QJE, Harel, Mossel, Strack, and Tamuz (2021). *Rational Groupthink* presents Theorem 1 as its first main result and Corollary 1 as a direct consequence.

ReStud, Farboodi, Jarosch, and Shimer (2023). *The Emergence of Market Structure* uses Lemma 1 on the way to Propositions 1 and 2, the section's main characterization results.

Takeaway. Lemma and corollary usually describe a result's relation to other results; theorem versus proposition is mainly a presentational choice.

Sources: Kamenica and Gentzkow, AER; Harel et al., QJE; Farboodi et al., ReStud.

How mathematical results are organized

Reading a result dependency

A COMMON DEPENDENCY PATTERN

- 1 Definitions & assumptions
- 2 Auxiliary lemmas
- 3 Theorems or propositions
- 4 Corollaries

COMMON ABBREVIATIONS

- | | |
|---------------|-----------------------|
| s.t. | such that; subject to |
| i.e. | that is |
| e.g. | for example |
| w.r.t. | with respect to |

Name results by their role

Assigning result labels

Results A , B , and C satisfy the following dependency structure.

A is proved only to establish B .

B is the conclusion of the argument.

C follows immediately from B .

MATCH THE ROLES

Match A , B , and C with

lemma,
theorem/proposition, and
corollary.

Names from the dependency structure

Labels implied by the dependency

- A* A **lemma**, because it is established to prove another result.
- B* A **theorem or proposition**, because it carries the conclusion of the argument.
- C* A **corollary**, because it follows immediately from *B*.